$$P(|E_{in} - E_{out}| > \varepsilon) \leq 2Me^{-2\varepsilon^2 N}$$

$$E_{out}(g) \leq E_{in}(g) + \sqrt{\frac{1}{2N} \ln \frac{2M}{\delta}}$$

If H is infinite, then $M \to \infty$, we have to replace M to be finite.

We are trying to band

$$|E_{in}(g) - E_{out}(g)| > \varepsilon$$

where the hypothesis g has a large gap btw training and test erors

B_m = Bad event = "$|E_{in}(h_m) - E_{out}(h_m)| > \varepsilon$"

$g \in H$, $H = \{h_1, h_2 \cdots h_m\}$

g must be contained in the event B_1 or B_2 or $\cdots B_M$

atleast one of hypothesis must be g

$\Rightarrow P(B_1 \cup B_2 \cdots \cup B_M)$

Union Bound says

$P(B_1 \text{ or } B_2 \text{ or} \cdots B_M) \leq$

$P(B_1) + P(B_2) + \cdots P(B_M)$

$$P(B_1 \cup \cdots B_M) \leq \sum_{m=1}^{M} P(B_m)$$

$$P(B_m) \leq 2e^{-2\varepsilon^2 N}$$

$$P(B_1 \cup \cdots B_M) \leq M \cdot 2e^{-2\varepsilon^2 N}$$

So this is where M comes from,

if events B_1, B_2, B_M are Strongly overlapping Union bound becomes loose.

if the bad events are highly correlated/overlapping (h_1 is very similar to h_2) $\Rightarrow |E_{in}(h_1) - E_{out}(h_1)| > \varepsilon$

and $|E_{in}(h_2) - E_{out}(h_2)| > \varepsilon$

conicides, then adding all their probablities overestimate the true probability.

Once we account for the overlaps of different hypothesis, we can replace M with an effective number that is finite

More overlap $\Rightarrow$ Union Bound becomes Looser